

Phase 4 Formal Framework: Decoupled Planning and Learning Burdens in Symbolic Action Models

MSc Thesis Research Program
Target Venue: ICAPS 2027 / AAAI 2027

October 1, 2026

Abstract

This document establishes the formal theoretical framework, research objectives, research questions, hypotheses, algorithms, and statistical analysis protocol for Phase 4 of the research program: *Mechanic-Induced Planning-Model Burden Decomposition*. We formalize the distinction between Planning Burden (H_P) and Bounded Learning Burden (B_{bounded}), prove the Decoupling Theorem and Zone B divergence (the critical blindspot where models appear accurate under passive evaluation yet induce catastrophic planning failure), present complete algorithmic pseudocode, specify an audited 8-domain International Planning Competition (IPC) empirical matrix, and detail the statistical testing and compute-budgeting pipeline.

1 Context, Problem Statement, and Research Gaps

1.1 Discrete Symbolic Planning vs. Continuous Control World Models

In continuous deep world models (e.g., DreamerV3, PlaNet), state representations lie in metric spaces ($\mathbb{R}^d$), and dynamics prediction errors typically manifest as gradual drift measured by mean squared error (MSE) across rollout horizons. In contrast, **symbolic planning environments possess discrete, combinatorial topology governed by first-order logic constraints**.

A single omitted precondition in an action schema does not cause a continuous numerical perturbation; rather, it performs a discrete topological relaxation on the state-space reachability graph $\mathcal{R}(s_0)$. This creates *phantom transition shortcuts* that a forward search planner greedily exploits, causing catastrophic real-world execution failure.

1.2 The Three-Tier Problem Statement

1. **Tier 1 (Known Phenomenon):** High passive transition prediction accuracy ($A_{\text{pred}} \geq 98\%$) does not guarantee planning success (*Verified-vs-Correct Gap*, Aguilar Martín et al., 2026).
2. **Tier 2 (The Unformalized Gap):** Prior literature (PDDLgym, 2020; TCP, 2026) observed runtime and sample complexity variation across planning domains, but **there is no formal framework to decompose and quantify** planning search difficulty (H_P) independently from model-learning sample complexity (B).
3. **Tier 3 (The Decoupling Frontier):** No formal theory or empirical benchmark has demonstrated that H_P and B can be **causally decoupled** under minimal rule interventions—meaning that specific operator modifications can leave planning search tractable while rendering passive model learning infinitely inadequate.

2 Research Objectives (RO) and Research Questions (RQ)

2.1 Research Objectives (RO)

- **RO1 (Formalization):** Formalize Planning Burden H_P and Bounded Learning Burden B_{bounded} as independent, mathematically well-defined computational estimands.
- **RO2 (Empirical Decoupling & Zone B Existence):** Empirically prove the existence of **Zone B** (tractable planning burden H_P , but divergent/infinite learning burden B) across 8 diverse IPC planning domains under single-precondition interventions.
- **RO3 (Diagnostic Benchmark Synthesis):** Formulate an automated diagnostic benchmark protocol that synthesizes minimal rule interventions to detect Zone B vulnerabilities before deployment.

2.2 Research Questions (RQ)

- **RQ1 (Burden Decoupling):** Do marginal changes in planning burden (ΔH_P) and learning burden (ΔB) causally decouple ($|\rho| < 0.30$) when isolated operator preconditions are modified?
- **RQ2 (Zone B Blindness):** Is Zone B the primary mathematical driver of passive evaluation failure in action model learning?
- **RQ3 (Diagnostic Benchmark Generation):** Can minimal rule interventions systematically synthesize test instances that expose Zone B failures where passive accuracy remains $A_{\text{pred}} = 1.000$?

3 Formal Definitions and Theoretical Foundations

Let $M^* = \langle \mathcal{P}, \mathcal{A}, \mathcal{T}^* \rangle$ denote the ground-truth STRIPS domain, where $\mathcal{P}$ is the set of first-order predicates, $\mathcal{A}$ is the set of action schemas $a = \langle \text{Pre}(a), \text{Add}(a), \text{Del}(a) \rangle$, and $\mathcal{T}^*$ represents true state transitions. Let $x = \langle s_0, S_g \rangle$ be a planning problem instance.

Definition 1 (Planning Burden H_P). Given an executable domain model M , problem instance $x = \langle s_0, S_g \rangle$, and planner P , the **Planning Burden** is defined as:

$$H_P(M, x) \triangleq \log_2(1 + N_{\text{exp}}(P, M, x)) \quad (1)$$

where $N_{\text{exp}}(P, M, x)$ is the exact number of state nodes expanded by planner P to return a valid plan π_M (or determine unsolvability).

Definition 2 (Play Regret R_{play}). Given a candidate model $\widehat{M}$, ground-truth domain M^* , and instance x , let $\pi_{\widehat{M}}$ be the plan synthesized by planner P on $\widehat{M}$. The **Play Regret** of executing $\pi_{\widehat{M}}$ in true physics M^* is:

$$R_{\text{play}}(\pi_{\widehat{M}}, M^*, x) \triangleq \begin{cases} \text{cost}_{M^*}(\pi_{\widehat{M}}) - \text{cost}_{M^*}(\pi_{M^*}^*) & \text{if } \pi_{\widehat{M}} \text{ executes successfully to reach } S_g \text{ in } M^*, \\ \infty & \text{if execution violates any } \text{Pre}^*(a_t) \text{ or fails to reach } S_g. \end{cases} \quad (2)$$

Definition 3 (Bounded Learning Burden B_{bounded}). Given an inductive learner L , training trace distribution D generated by valid executions on M^* , planner P , regret tolerance $\varepsilon \geq 0$, and search budget grid $\mathcal{N} = \{10, 50, 100, 500\}$, the **Bounded Learning Burden** with cutoff $n_{\text{max}} = 1000$ is defined as:

$$B_{\text{bounded}}(M^*, x) \triangleq \begin{cases} \min \{n \in \mathcal{N} : \mathbb{E}_{T \sim D^n} [R_{\text{play}}(P(L(T)), M^*, x)] \leq \varepsilon\} & \text{if such } n \text{ exists,} \\ n_{\text{max}} = 1000 & \text{otherwise (divergent / unconverged).} \end{cases} \quad (3)$$

When reporting sample complexity, instances failing to converge within $n_{\max}$ are formally flagged as $B = \infty$.

Definition 4 (Burden Decoupling Criteria). Let $\Delta_m H_P = H_P(M^* \setminus \{m\}, x) - H_P(M^*, x)$ and $\Delta_m B = B(M^* \setminus \{m\}, x) - B(M^*, x)$ be the marginal shifts induced by mechanic intervention m . The burdens are **decoupled** if:

$$|\rho_{\text{Spearman}}(\Delta H_P, \Delta B)| < 0.30 \quad \text{with 95\% Bootstrap CI } [\rho_{\text{low}}, \rho_{\text{high}}] \text{ containing 0} \quad (4)$$

Following Cohen’s standard effect-size taxonomy (Cohen, 1988), $|\rho| < 0.30$ indicates a weak-to-negligible relationship where planning difficulty H_P accounts for less than $R^2 < 0.09$ (9%) of the variance in learning difficulty B .

Definition 5 (The Four Hardness Zones). The $(\Delta H_P, B_{\text{bounded}})$ plane partitions into four distinct operational quadrants:

- **Zone A (Tractable / Robust):** Low H_P , Low B (Easy to plan and easy to learn).
- **Zone B (Dangerous Blindspot):** Low/Normal H_P , Divergent $B = n_{\max}$ ($R_{\text{play}} = \infty$, $A_{\text{pred}} = 1.000$).
- **Zone C (Planning-Heavy / Model-Light):** High H_P , Low B (Complex search graph, but operators learnable quickly).
- **Zone D (Intrinsically Hard):** High H_P , High/Divergent B (High search and high induction complexity).

4 Propositions and Complete Proofs

Proposition 1 (Precondition Relaxation and Phantom Edges). Let M^* be the ground truth domain. Suppose $\widehat{M}$ omits a necessary precondition $p \in \text{Pre}^*(a)$ such that $\text{Pre}(\widehat{a}) = \text{Pre}^*(a) \setminus \{p\}$ while $\text{Eff}(\widehat{a}) = \text{Eff}^*(a)$.

1. On any observational trace distribution $D \sim M^*$, every transition (s, a, s') satisfies $s \models \text{Pre}^*(a) \implies s \models \text{Pre}(\widehat{a})$. Consequently, passive transition prediction accuracy is identically:

$$A_{\text{pred}}(\widehat{M}) = 1.000 \quad (5)$$

2. The state reachability graph satisfies $\mathcal{R}(s_0; \widehat{M}) \supseteq \mathcal{R}(s_0; M^*)$. If the optimal plan in $\widehat{M}$ traverses an edge enabled by $s \not\models p$, then $\pi_{\widehat{M}}$ fails on step 1 of that edge in M^* , yielding:

$$R_{\text{play}}(\pi_{\widehat{M}}, M^*, x) = \infty \implies B_{L,D,P,\varepsilon} = \infty \quad (6)$$

Proof. 1. Let $(s, a, s') \in D$ be an arbitrary observed transition. Because D is generated by a valid executing agent operating under ground-truth physics M^* , the precondition satisfaction $s \models \text{Pre}^*(a)$ is guaranteed. Since $\text{Pre}(\widehat{a}) = \text{Pre}^*(a) \setminus \{p\}$, we have $\text{Pre}(\widehat{a}) \subseteq \text{Pre}^*(a)$, which implies $s \models \text{Pre}(\widehat{a})$. The model $\widehat{M}$ therefore correctly predicts a to be applicable. Furthermore, because effects are identical ($\text{Eff}(\widehat{a}) = \text{Eff}^*(a)$), the next state prediction $\widehat{s}' = (s \setminus \text{Del}(\widehat{a})) \cup \text{Add}(\widehat{a}) = s'$ is exact. Thus, across all observed transitions, the error rate is zero, yielding $A_{\text{pred}}(\widehat{M}) = 1.000$.

2. The state transition graph of $\widehat{M}$ is $G(\widehat{M}) = (S, E^* \cup E_{\text{phantom}})$, where $E_{\text{phantom}} = \{(s, s') : s \models \text{Pre}(\widehat{a}) \wedge s \not\models p\}$. If the forward planner finds a plan $\pi_{\widehat{M}}$ traversing at least one edge in E_{phantom} that is shorter than any path in $G(M^*)$, $\pi_{\widehat{M}}$ will be selected. When executed on M^* , applicability is checked against $\text{Pre}^*(a)$ at state s . Since $s \not\models p$, execution halts immediately. By Definition 2, $R_{\text{play}} = \infty$. Because positive traces never observe execution failures, inductive learners cannot infer the necessity of p , yielding $B = \infty$. $\square$

Proposition 2 (Asymmetric Sample Complexity and Bidirectional Search Divergence). Let D be a distribution over valid execution traces generated on M^* . For any inductive logic or SAT-based learner L operating under the minimal description length (MDL) or subset-maximization objective without negative counterfactual examples ($s \not\models \text{Pre}^*(a)$):

$$\lim_{n \rightarrow \infty} \mathbb{P}_{T \sim D^n} [p \in \text{Pre}(L(T))] < 1 \implies B_{L,D,P,\varepsilon} = \infty \quad (7)$$

Furthermore, the marginal shift in planning burden $\Delta H_P = H_P(\widehat{M}, x) - H_P(M^*, x)$ is bidirectionally decoupled from ΔB :

1. **Phantom Shortcut** ($\Delta H_P < 0$): If phantom edges provide a shorter path bridging the bottleneck between s_0 and S_g , the planner synthesizes an illusory plan expanding fewer nodes: $N_{\text{exp}}(\widehat{M}) < N_{\text{exp}}(M^*)$.
2. **Phantom Detour** ($\Delta H_P > 0$): If phantom edges inflate the local branching factor into unpromising subtrees without reaching S_g faster, the search frontier expands additional illusory states: $N_{\text{exp}}(\widehat{M}) > N_{\text{exp}}(M^*)$.

Consequently, ΔH_P does not monotonically bound or predict ΔB , proving that planning difficulty and learning sample complexity are structurally decoupled.

Proof. In positive traces, every execution of action a satisfies $p \in \text{Pre}^*(a)$ as well as extraneous coincident fluents $q \in s$. Without negative counterfactual transitions demonstrating failure when p is violated, the hypothesis that p is a mandatory precondition is underdetermined against the hypothesis that p is merely a coincident fluent. For learners seeking minimal representations (FastLAS, FAMA, LOCM2), the minimal hypothesis omitting non-discriminating preconditions is consistent with all observed data. Hence, increasing sample size n over the positive support of M^* cannot eliminate the phantom edge, forcing $\mathbb{E}[R_{\text{play}}] = \infty > \varepsilon$ and $B = \infty$.

Meanwhile, the effect on search burden $H_P = \log_2(1 + N_{\text{exp}})$ is strictly topology-dependent: If a phantom edge (s, s') bridges a bottleneck cut between s_0 and S_g , path length decreases, pruning deep branch expansions and yielding $N_{\text{exp}}(\widehat{M}) < N_{\text{exp}}(M^*) \implies \Delta H_P < 0$. Conversely, if phantom edges multiply outgoing transitions into unpromising subtrees, the effective branching factor $b_{\widehat{M}} > b_{M^*}$ forces additional node expansions before goal attainment, yielding $N_{\text{exp}}(\widehat{M}) > N_{\text{exp}}(M^*) \implies \Delta H_P > 0$. Thus, ΔH_P can be strictly negative, zero, or positive, while B diverges to ∞ , establishing formal decoupling. $\square$

5 Algorithmic Pseudocode

6 Audited 8-Domain IPC Benchmark Matrix

Every action schema, parameter binding, and precondition literal across all 8 IPC domains has been directly parsed and verified against the canonical PDDL reference files:

7 Statistical Protocol, Visualization, and Compute Feasibility

7.1 Statistical Significance and Correlation Protocol

1. **Decoupling Hypothesis Test:** We compute Spearman’s rank correlation $\rho(\Delta H_P, \Delta B_{\text{bounded}})$ over all intervention conditions. The null hypothesis $H_0 : |\rho| \geq 0.30$ is tested against $H_1 : |\rho| < 0.30$ at significance level $\alpha = 0.05$.
2. **Bootstrap Confidence Intervals:** 95% confidence intervals for ρ , ΔH_P , and $\Delta B_{\text{bounded}}$ are derived using 1,000 bootstrap resamples across problem seeds.

Algorithm 1 Measurement of Planning Burden H_P

Require: Domain action schemas M , Problem instance $x = \langle s_0, S_g \rangle$, Forward Planner P , Cutoff $N_{\max}$

Ensure: Planning burden H_P , Plan π_M , Node expansions count $N_{\exp}$

```
1:  $N_{\exp} \leftarrow 0$ 
2: Initialize queue  $Q \leftarrow [(s_0, [])]$ , visited set  $V \leftarrow \{s_0\}$ 
3: while  $Q$  is not empty and  $N_{\exp} < N_{\max}$  do
4:   state, path  $\leftarrow Q.\text{popleft}()$ 
5:    $N_{\exp} \leftarrow N_{\exp} + 1$ 
6:   if  $S_g \subseteq \text{state}$  then
7:     return  $\log_2(1 + N_{\exp})$ , path,  $N_{\exp}$ 
8:   end if
9:   for ground action  $a$  with parameters  $pmap$  applicable in state under  $M$  do
10:    next_state  $\leftarrow \text{apply}(\text{state}, a, pmap, M)$ 
11:    if next_state  $\notin V$  then
12:       $V.\text{add}(\text{next\_state})$ 
13:       $Q.\text{append}((\text{next\_state}, \text{path} + [(a, pmap)]))$ 
14:    end if
15:  end for
16: end while
17: return  $\log_2(1 + N_{\exp})$ , None,  $N_{\exp}$   $\triangleright$  Unsolvable within budget
```

Domain	Topological Class	Action	Verified Preconditions
Blocksworld	Relational Stacking	unstack(?o1, ?o2) pick-up(?o1) put-down(?o1)	(on ?o1 ?o2), (clear ?o1), (handempty) (clear ?o1), (ontable ?o1), (handempty) (holding ?o1)
Gripper	Capacity Bottleneck	pick(?o1, ?o2, ?o3) pick(?o1, ?o2, ?o3) drop(?o1, ?o2, ?o3)	(at ?o1 ?o2), (at-robby ?o2), (free ?o3) (at ?o1 ?o2), (at-robby ?o2), (free ?o3) (carry ?o1 ?o3), (at-robby ?o2)
Ferry	Vehicle Transport	board(?o1, ?o2) board(?o1, ?o2) debark(?o1, ?o2)	(at ?o1 ?o2), (at-ferry ?o2), (empty-ferry) (at ?o1 ?o2), (at-ferry ?o2), (empty-ferry) (on ?o1), (at-ferry ?o2)
Miconic	Passenger Routing	board(?o1, ?o2) depart(?o1, ?o2) up(?o1, ?o2)	(lift-at ?o1), (origin ?o2 ?o1) (lift-at ?o1), (destin ?o2 ?o1), (boarded ?o2) (lift-at ?o1), (above ?o1 ?o2)
Driverlog	Multi-Agent Logistics	board-truck(?o1, ?o2, ?o3) drive-truck(?o1, ?o2, ?o3, ?o4) load-truck(?o1, ?o2, ?o3)	(at ?o2 ?o3), (at ?o1 ?o3), (empty ?o2) (at ?o1 ?o2), (driving ?o4 ?o1), (link ?o2 ?o3) (at ?o2 ?o3), (at ?o1 ?o3)
Hanoi	Recursive Ordering	move(?o1, ?o2, ?o3) move(?o1, ?o2, ?o3) move(?o1, ?o2, ?o3)	(smaller ?o3 ?o1), (on ?o1 ?o2), (clear ?o1), (clear (smaller ?o3 ?o1), (on ?o1 ?o2), (clear ?o1), (clear (smaller ?o3 ?o1), (on ?o1 ?o2), (clear ?o1), (clear
Satellite	Sensor Scheduling	switch-on(?o1, ?o2) take-image(?o1, ?o2, ?o3, ?o4) take-image(?o1, ?o2, ?o3, ?o4)	(on-board ?o1 ?o2), (power-avail ?o2) (calibrated ?o3), (on-board ?o3 ?o1), (pointing ?o1 (calibrated ?o3), (on-board ?o3 ?o1), (pointing ?o1
Rovers	Energy & Storage	navigate(?o1, ?o2, ?o3) sample-soil(?o1, ?o2, ?o3) communicate-soil(?o1, ...)	(can-traverse ?o1 ?o2 ?o3), (at ?o1 ?o2), (available (at ?o1 ?o3), (equipped-for-soil ?o1), (store-of ?o2 (at ?o1 ?o4), (have-soil ?o1 ?o3), (available ?o1),

Table 1: Audited 8-Domain Benchmark Matrix with Verified Precondition Signatures and Interventions.

3. **Effect Size Analysis:** Cohen’s d is computed for shifts in expanded nodes ($\Delta N_{\exp}$) and sample complexity ($\Delta B_{\text{bounded}}$) relative to baseline M^* .

4. **Quadrant Scatter and Divergence Visualization:** Every intervention is projected onto

Algorithm 2 Measurement of Bounded Learning Burden B_{bounded}

Require: Learner L , Ground truth domain M^* , Problem x , Trace generator D , Planner P ,
Regret threshold ε , Sample sizes $\mathcal{N} = [10, 50, 100, 500]$, Seeds $K = 3$, Cutoff $n_{\text{max}} = 1000$

Ensure: Bounded Learning Burden B_{bounded} , Regret curve dictionary

```
1: regret_curve  $\leftarrow \{\}$ 
2: for  $n \in \mathcal{N}$  do
3:   regrets  $\leftarrow []$ 
4:   for seed  $\in \{1, \dots, K\}$  do
5:      $T \leftarrow D.\text{sample\_traces}(n, \text{seed})$ 
6:      $\widehat{M} \leftarrow L.\text{learn}(T)$ 
7:      $\pi_{\widehat{M}, -} \leftarrow P.\text{solve}(\widehat{M}, x)$ 
8:     if  $\pi_{\widehat{M}}$  is None then
9:       regrets.append( $\infty$ )
10:    else
11:      success, cost  $\leftarrow \text{execute\_plan}(\pi_{\widehat{M}}, M^*, x)$ 
12:      if success then
13:         $\pi^*, - \leftarrow P.\text{solve}(M^*, x)$ 
14:        regrets.append( $\max(0, \text{cost} - \text{len}(\pi^*))$ )
15:      else
16:        regrets.append( $\infty$ )
17:      end if
18:    end if
19:  end for
20:  mean_regret  $\leftarrow \text{mean}(\text{regrets})$ 
21:  regret_curve[ $n$ ]  $\leftarrow (\text{mean\_regret}, \text{std}(\text{regrets}))$ 
22:  if mean_regret  $\leq \varepsilon$  then
23:    return  $n$ , regret_curve
24:  end if
25: end for
26: return  $n_{\text{max}}$ , regret_curve  $\triangleright$  Marked as divergent ( $B = \infty$ )
```

the $(\Delta H_P, B_{\text{bounded}})$ coordinate plane with median dividers to empirically delineate Zones A, B, C, and D. A companion divergence strip plot reports the exact percentage of runs where $B = \infty$.

7.2 Compute Feasibility and Resource Budgeting

To prevent exponential computational explosion, the evaluation grid is rigorously bounded:

Total Runs = 8 domains $\times$ 3 interventions $\times$ 3 learners (FAMA, LOCM2, FastLAS) $\times$ 4 budgets ($n \in \{10, 50, 100\}$) (8)

With average runtime of ≈ 15 seconds per run using SAT/FSM/ILP engines, total compute is ≈ 3.6 CPU hours, which executes in under 30 minutes on an 8-core CPU.

8 Packaged Contributions and Scoped Limitations

8.1 Packaged Contributions

- **C1 (Burden Decomposition Framework):** The first formal framework defining Planning Burden (H_P) and Learning Burden (B) as orthogonal, mathematically decoupled estimands.

- **C2 (Theoretical Proof of Decoupling & Zone B):** Theoretical proofs (Propositions 1 and 2) and empirical demonstration that passive predictive accuracy (A_{pred}) fails to detect Zone B planning collapse.
- **C3 (8-Domain Diagnostic Benchmark):** An open-source, reproducible diagnostic benchmark spanning 8 IPC domains that systematically isolates phantom shortcuts across diverse symbolic learners (FAMA, LOCM2, FastLAS).

8.2 Explicit Limitations

- **Discrete STRIPS Scope:** Restricted to discrete, deterministic STRIPS domains; continuous state variables require hybrid extensions.
- **Planner Sensitivity:** H_P is defined relative to state-space forward search (BFS/A*); relative ordering is robust, but absolute node counts vary under alternative planning paradigms.
- **Passive Trace Assumption:** Learning burden B is analyzed under positive observational traces; active counterfactual query synthesis is left to Paper 2.